



IDENTIFICATION

CODE : GI-3-S2-EC-CDA
ECTS : 3

HOURS

Cours : 0h
TD : 26h
TP : 20h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 48h
Travail personnel : 12h
Total : 60h

ASSESSMENT METHOD

ES (évaluation en situation)
ES (évaluation en situation)
ES (évaluation en situation)
ES (évaluation en situation)

TEACHING AIDS

The pedagogic form will insist on the transcription of analytical expression to plant reality.
Support are handout and annales

TEACHING LANGUAGE

French

CONTACT

MME SUBAI Corinne :
corinne.subai@insa-lyon.fr

AIMS

This course belongs to teaching unit Pilotage des systèmes industriels (GI-3-S2-UE-PI(SI)) and contributes to the following skills :

- C2 Modeling and designing an information, decision and production system, of goods and services (Level 3)
- C3 Evaluating, prototyping and simulating a system (Level 3)
- C4 Sizing the hardware and / or software of a system (Level 3)
- C8 Managing supply in connection with the planning and inventory management policy (Level 3)
- C9 Localizing and assigning the production, storage and transportation processes to different members of the supply chain (Level 3)
- C17 Identifying, formalizing and contractualising the requirements of a client, following their evolution and validating their compliance (requirements traceability) (Level 3)
- B2 Work, learn, progress autonomously (Level 3)

By allowing the engineering students to work and be evaluated on the following knowledge :

- Modeling and designing an information, decision and production system, of goods and services (C2)
- Evaluating, prototyping and simulating a system (C3)
- Managing supply in connection with the planning and inventory management policy (C8)

To be able to :

- Identifying, formalizing and contractualising the requirements of a client, following their evolution and validating their compliance (requirements traceability) (C17)
- Sizing the hardware and / or software of a system (C4)
- Localizing and assigning the production, storage and transportation processes to different members of the supply chain (C9)
- Work, learn, progress autonomously (B2)

CONTENT

Part 1 : Petri Nets (PN)

- 1.1 Introduction to the resource/flow paradigm
- 1.2 Autonomous PN (definition, properties)
- 1.3 Non-autonomous PN (synchronised, temporal)
- 1.4 Analysis (dimensionning and performance assessment)

Partie 2 : Queueing Nets (QN)

- 2.1 Introduction to the customer/server paradigm
- 2.1 Markov Chains and state equation
- 2.2 Performance indicators
- 2.3 Conventional Station
- 2.4 Markov Net limited to product form

BIBLIOGRAPHY

Du graficet aux réseaux de Petri, R. David, H. Alla HERMES 1989.
Modélisation des systèmes de production, J.M. Proth, X. Xie DUNOD 1992.
Introduction to DES, M. Cassandras, S. Lafortune, Wiley et sons 1999.
Modelling with generalized stochastic Petri Nets, A. Marsan, S. Donatelli Wiley et sons Ltd 1995.
Processus stochastiques, Alan Ruegg, méthodes mathématiques pour l'ingénieur, presses polytechniques romandes, Lausanne, 1989.
Network flow programming, by P. A. Jensen and J. W. Barnes, John Wiley and Sons, Inc., New York, 1980.

PRE-REQUISITES

Bachelor Eng.
Automatic Control
Discrete Mathematic
Graph Theory
Probability and Statistics

